

Bone marrow stem cell connexins: misconceptions and new insights

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Hematopoietic regeneration requires coordinated activation of hematopoietic stem and progenitor cells (HSPCs) and adaptive remodeling of the bone marrow microenvironment to meet extreme metabolic and oxidative demands imposed by cytotoxic injury, transplantation, and inflammation. Although soluble factors and cytokine signaling are central to this process, emerging evidence identifies direct intercellular communication as a critical regulatory layer in stress hematopoiesis. Connexins, particularly connexin-43 (Cx43), form an evolutionarily conserved communication network that integrates metabolic coupling, redox buffering, and organelle dynamics across hematopoietic and stromal compartments. Beyond canonical gap junction channel activity, connexins exert nonjunctional, compartment-specific functions through cytoplasmic, nuclear, and mitochondrial pools that regulate signaling scaffolds, transcriptional programs, cytoskeletal organization, mitochondrial dynamics, calcium homeostasis, and bioenergetics. In HSPCs, mitochondrial Cx43 functions as a metabolic checkpoint that preserves

regenerative capacity by supporting oxidative phosphorylation, limiting chronic adenosine monophosphate-activated protein kinase activation, maintaining fusion-fission balance, and preventing mitochondrial Ca²⁺ overload. In parallel, Cx43 enables mitochondrial transfer from donor HSPCs to stromal niche cells, restoring stromal metabolic competence and promoting effective niche repair and engraftment. Dysregulation of connexin networks contributes to marrow failure, clonal evolution, leukemic niche remodeling, and chemoresistance, highlighting their context-dependent roles in health and disease. This review synthesizes advances in connexin biology in hematopoiesis, reframes connexins as integrators of metabolic and regenerative signaling rather than passive conduits, and defines emerging translational opportunities. Isoform- and compartment-specific targeting of connexin pathways offers a therapeutic strategy to enhance hematopoietic recovery, preserve long-term stem cell function, and disrupt pathological niche support in hematologic malignancies.

Introduction

Hematopoietic stem cells (HSCs) sustain lifelong blood production through tightly regulated programs of self-renewal, proliferation, and differentiation.¹ Under homeostasis, HSCs remain largely quiescent to preserve long-term function.² In contrast, regeneration after chemotherapy, irradiation, or transplantation exposes hematopoietic stem and progenitor cells (HSPCs) to extreme metabolic and oxidative stress.³ Effective recovery requires rapid progenitor expansion, preservation of stem cell fitness, and coordinated remodeling of the bone marrow (BM) microenvironment.⁴

Once thought to cycle uniformly, the HSC compartment is now recognized as heterogeneous, comprising subpopulations with distinct propensities to enter the cell cycle.⁵ This diversity generates reserve pools that can be selectively mobilized to meet regenerative demands across stresses and life span.⁶

During regeneration, HSCs must exit quiescence and proliferate, a necessary but hazardous transition that increases bioenergetic demand and vulnerability to DNA damage, oxidative stress, and functional exhaustion.⁷ Therefore, successful regeneration depends on molecular programs that support proliferation with high fidelity while preserving long-term self-renewal.⁸ Despite significant efforts to identify the metabolic, signaling, and organellar mechanisms distinguishing quiescent from cycling HSCs,⁹⁻¹¹ these mechanisms remain incompletely defined.

Soluble factors such as cytokines and growth factors, central to hematopoietic regeneration, do not fully explain how HSPCs meet increased bioenergetic demands under stress, buffer oxidative stress, or how injured niches restore metabolic competence.¹² Emerging evidence highlights direct cell-to-cell communication as a critical, underappreciated regulatory layer during stress hematopoiesis.¹³⁻¹⁸ In metazoans, direct

cell-to-cell communication is achieved by gap junctions (GJs), cell membrane junctional hexagonal structures that display a 20 to 30 Å “gap” between adjacent cells.¹⁹ In chordates, the proteins that are involved in the formation of GJs are called connexins.²⁰

Connexins constitute an evolutionarily conserved family of 200 to 500 amino acid proteins, with 21 members in humans and 20 in mice, classified into 5 subfamilies (α , β , γ , δ , and ϵ).²¹ Although most mammalian cells express ≥ 1 connexin, transcript abundance does not necessarily predict functional GJ assembly or channel activity.²² In BM, connexin-43 (Cx43), a 43-kDa protein encoded by *GJA1*, is the predominant connexin expressed in both hematopoietic and stromal compartments, with additional expression of Cx37, Cx40, and Cx45, in adult BM stromal compartments that include endothelial, mesenchymal, and osteolineage cells; Cx31, in neonatal BM; and Cx30.3 and Cx31.1, in fetal liver hematopoietic-supporting stroma.¹³ This combinatorial expression pattern is expected to fine-tune the regulation of intercellular communication during steady-state and stress hematopoiesis.

Connexins function through docking-dependent channels²³ or unopposed hemichannels.²⁴ Docked GJ channels permit the direct transfer of ions and small metabolites up to ~1.2 kDa, including second messengers and metabolic intermediates, synchronizing signaling and metabolism between neighboring cells.^{25–28} Undocked connexin hemichannels (also called connexons) mediate the regulated extracellular release of signaling molecules such as ATP and NAD^+ , influencing autocrine and paracrine signaling.^{28–30} The connexin composition of the channels crucially controls the permeability of its internal pore with different permeability capabilities for negatively vs positively charged molecules and molecular size.³¹ Beyond channel activity, connexins (or alternatively expressed C-terminus containing isoforms)^{32,33} may also exert noncanonical, channel-independent functions through interactions of their cytoplasmic C-terminal domains with signaling³⁴ and cytoskeletal proteins,³⁵ regulating mitochondrial dynamics¹⁸ and cation permeability,^{18,36} cell polarity,³⁷ and stress-responsive signaling pathways.^{38,39}

Cx43 is highly expressed in cycling HSCs and early progenitors, declining during differentiation.^{14–16,40,41} Initially implicated in niche retention and migration,^{14,42} Cx43 is now recognized as a key regulator of HSC metabolism and stromal support. Mitochondrial pools of Cx43 stabilize membrane potential, optimize oxidative phosphorylation (OXPHOS), and buffer reactive oxygen species (ROS), supporting progenitor fitness during regenerative stress.^{16,18,43} Connexin-dependent signaling also facilitates intercellular metabolic support through indirect mitochondrial transfer from HSPCs to BM stromal cells, restoring niche metabolism and hematopoietic support factor production.^{16,18}

This review summarizes advances in connexin biology of BM hematopoiesis. We discuss how connexin-dependent metabolic coupling and mitochondrial regulation shape hematopoietic regeneration, leukemic niche interactions, and therapeutic resistance, highlighting emerging opportunities to target connexin pathways to enhance engraftment, accelerate recovery, and preserve long-term stem cell function.

Connexin-mediated intercellular communication in the HSC niche

Non-HSPC autonomous roles of Cx43 in the BM niche(s)

Specific cellular interactions can be dissected in vivo using cell-type-specific genetic mutant models, which enable precise functional interrogation of intercellular signaling and niche regulation within intact tissues. Effective hematopoietic recovery requires coordinated repair of hematopoietic cells and the surrounding microenvironment, using connexins to exert cell-autonomous and noncell-autonomous effects on HSC activity.¹⁵ Effective engraftment of HSCs requires migration and lodging in proper niche sites. Considerable evidence has demonstrated that BM mesenchymal stromal cells (MSC), osteoblasts, mesenchymal stem cells, and endothelial cells are extensively coupled by GJs.^{13,14,44} Evidence from conditional deletion of Cx43 in the BM HSC niche shows that niche connexin expression is critical for directional homing and mobilization under stress.^{14,44}

MSCs and osteolineage cells MSCs and osteolineage populations express multiple connexins, with Cx43 being highly expressed in osteoblasts and osteocytes.^{45–47} Connexin signaling in these compartments regulates lineage commitment (eg, osteogenesis vs adipogenesis),^{48,49} mechanical responsiveness,^{50,51} and the secretion of niche factors such as CXCL12 and PGE_2 .⁴² Through these mechanisms, connexin activity in the osteolineage compartment influences bone modeling/remodeling, in turn shaping marrow architecture and niche occupancy.^{14,46,52} For example, Cx43 in osteoblasts modulates PGE_2 /PTH signaling to regulate bone remodeling, and its deletion leads to abnormal ossification and skeletal defects.^{45,53}

Conditional deletion of Cx43 in osteogenic niche cells (osteoblast/progenitor compartment) reveals osteolineage connexin expression as dispensable for steady-state hematopoiesis but essential for directional homing and mobilization under regenerative stress.¹⁴ In myeloablated recipients, stromal Cx43 deficiency disrupts transstromal migration, impairing HSC homing and mobilization.¹⁴ Mechanistically, loss of osteolineage Cx43 increases CXCL12 expression and secretion, perturbing spatial chemokine gradients that guide marrow HSC positioning. This dysregulation propagates aberrant HSC retention near endosteal surfaces and reduces circulatory mobilization,¹⁴ supporting connexin-mediated coupling within the stroma as a critical determinant of spatial organization of chemokine cues governing HSC distribution, trafficking, and regeneration.

Vascular and perivascular niches BM endothelial cells express Cx37, Cx40, Cx43, and Cx45 and establish functional GJs with perivascular stromal elements, enabling ionic, metabolic, calcium, and ATP flux.^{24,28,54,55} These interactions are critical for HSC homing, leukocyte adhesion and transmigration, and stem cell retention within the BM. Connexin-mediated communication between endothelial and perivascular compartments integrates vascular tone, chemotactic signals such as CXCL12/CXCR4 and Rac/Rho GTPases, and intracellular signaling pathways to dynamically redistribute

HSPCs during both homeostasis and stress hematopoiesis.^{14,17,42,44,54} Loss of endothelial Cx43 during angiotensin II-driven hyperangiotensinemia alters cytoskeletal organization and RhoA signaling in HSPCs and endothelial cells, reduces adhesive interactions, and increases mobilization of long-term and short-term HSCs and progenitors into the circulation.⁴⁴ Therefore this coordinated coupling maintains HSCs within a metabolically supportive environment that can rapidly respond to regenerative demands.

Immune and macrophage interactions Cx43 is expressed in myeloid and lymphoid cells, enabling heterocellular coupling with stromal and endothelial compartments.^{54,56-59} These intercellular channels facilitate the transfer of small signaling molecules and metabolites, including Ca^{2+} , ATP, NAD^+ , inositol trisphosphate, and redox intermediates, allowing propagation of calcium, modulation of cytokine signaling, and coordination of metabolic states between immune and niche cells during inflammation.^{58,60-62} Myeloid-stromal connexin coupling regulates local chemokine production, ROS signaling, and NF- κ B-dependent inflammatory responses, reshaping the microenvironment governing HSC proliferation, differentiation, and mobilization under stress.^{17,63} Hence, connexin-mediated communication is an intercellular signaling network linking hematopoietic, immune, and stromal compartments, integrating inflammatory cues with regenerative hematopoietic programs.

Interplay between adipogenesis and hematopoietic niche functions Adipocytes, part of normal BM, act as positive regulators of normal and stress hematopoiesis.⁶⁴⁻⁶⁶ However, specific regionally distributed BM adipocytes may exert negative regulatory effects,⁶⁷ which may accentuate after aging-associated accumulation, reshaping the BM microenvironment and negatively influencing hematopoietic regeneration in mice and humans.^{68,69} Connexin signaling, particularly via Cx43, directs MSC fate by modulating adipogenic vs osteogenic differentiation, thereby determining the niche composition and function.^{70,71} By shifting the adipocyte-to-osteoblast ratio, connexin dynamics alter stromal support cues, such as CXCL12 and PGE_2 , ultimately affecting HSPC maintenance, differentiation, and regenerative potential.^{14,72}

Extracellular mediators and paracrine coupling Connexin channels extend niche communication beyond direct cell-to-cell coupling by regulating paracrine signaling. ATP and other small molecules released through hemichannels or GJs may act on purinergic receptors, trigger calcium signaling, or modulate prostaglandin, cytokine, and chemokine release.^{21,28,30,73} This coupling integrates biochemical feedback, enabling niche sensing and responses to metabolic and inflammatory cues. By modulating the local microenvironment, connexin activity can influence HSPC migration, stromal cell behavior, and extracellular matrix remodeling (eg, collagen deposition), coordinating hematopoietic homeostasis and regeneration through secondary messengers.^{24,42}

Cx43 in oxidative stress buffering and senescence prevention Cx43 mediates intercellular transfer of ROS from HSCs to BM stromal cells, buffering intracellular ROS levels in HSCs and preventing premature senescence.¹⁵ Hematopoietic-specific Cx43 deficiency delays myeloablation recovery, increases senescence marker (p16^{INK4a} and p21) expression, p38

MAPK activation, and produces oxidative stress-induced HSC dysfunction.¹⁵ Over time, Cx43-deficient mice exhibit accelerated functional exhaustion and aging-like phenotypes within HSC compartments. In vitro assays confirm that HSCs can offload excess ROS to stromal cells via Cx43-dependent GJ communication, whereas antioxidants and Cx43 reexpression restore HSC function.¹⁵ This positions Cx43 as a regulator of oxidative stress homeostasis, safeguarding against premature stem cell aging, particularly under stress.

Mitochondrial exchange as a regulator of niche regeneration and therapy response BM regeneration requires not only intrinsic progenitor fitness but also repairs of the stromal niche. Donor HSPC transfers mitochondria to stromal cells, restoring oxidative metabolism and hematopoietic support factor production, a process that requires Cx43 expression in donor HSPCs.¹⁶ Cx43 does not serve as a physical conduit for mitochondrial passage. Instead, it is a signaling hub that coordinates upstream processes such as mitochondrial network remodeling, Ca^{2+} flux, and metabolic sensing pathways (eg, adenosine monophosphate-activated protein kinase [AMPK]-dependent signaling). By integrating these signals, Cx43 promotes cytoskeletal polarization and intracellular trafficking programs that enable mitochondrial mobilization and redistribution.^{16,18}

Oxidative stress and metabolic scarcity promote intercellular mitochondrial trafficking through mechanisms including tunneling nanotubes (TNTs), clathrin-dependent uptake of extracellular vesicles, cell fusion, and mitochondrial extrusion, enhancing recipient cell bioenergetics and survival.^{43,74-78} TNTs represent a major route for mitochondrial transfer, and Cx43 stabilizes these structures by forming junctional plaques at contact sites, anchoring nanotubes, and preserving their mechanical integrity during organelle transport.⁷⁹⁻⁸¹ In addition to TNT-mediated transfer, Cx43 supports vesicle-mediated mitochondrial exchange, enabling longer-range paracrine communication.^{82,83} Intracellular mitochondrial trafficking is coordinated by the calcium-sensitive Rho GTPases Miro1 and Miro2, which regulate microtubule and actin-dependent transport cooperating with Cx43-dependent TNT formation to facilitate mitochondrial donation from MSCs to surrounding cells.^{43,75,77,78}

Cx43-mediated transfer enables delivery of functional mitochondria from BM MSCs to HSCs, supporting bioenergetic demands during inflammatory or injury-associated stress.⁷⁸ In regenerating BM, mitochondrial exchange is bidirectional but preferentially directed from hematopoietic cells to stromal MSCs, promoting repair of damaged niche cells and reflecting dynamic metabolic coupling within the microenvironment.¹⁶ Cx43 selectively regulates mitochondrial transfer from HSPCs to Sca1⁺PDGFR⁺ MSC subsets, with $\leq 90\%$ of MSCs acquiring donor-derived mitochondria within 1 month after transplantation.¹⁶ Transferred mitochondria remain structurally intact and functionally integrated, restoring membrane potential and increasing ROS production.¹⁶

Efficient transfer requires direct cell contact, preserved mitochondrial viability, and a metabolically active donor pool. In Cx43-deficient HSPCs, excessive mitochondrial fission and mitophagy reduce this transferable pool and markedly limit mitochondrial donation.¹⁸ ATP-P2RX7 signaling promotes

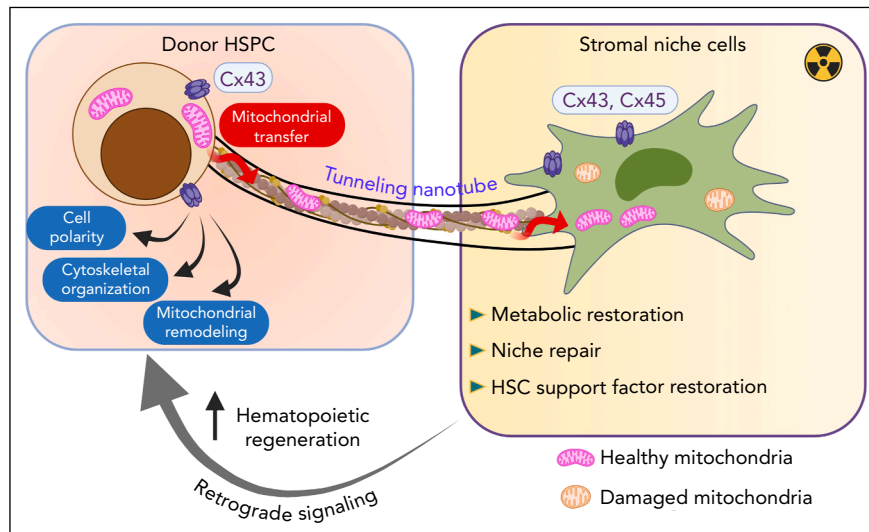


Figure 1. Cx43-dependent mitochondrial transfer supports stromal niche repair during hematopoietic regeneration. During BM regeneration following transplantation or cytotoxic injury, donor HSPCs transfer mitochondria to stromal niche cells, restoring stromal oxidative metabolism and hematopoietic support factor production. Cx43 expression in HSPCs is essential for this process, not as a physical conduit for organelle passage, but through regulation of upstream mechanisms, including mitochondrial network remodeling, cellular polarity, and cytoskeletal organization. Mitochondrial trafficking occurs through TNT-like structures, enabling reciprocal metabolic support between hematopoietic cells and the niche, thereby promoting effective engraftment and regenerative hematopoiesis. Figure created with BioRender.com. Singh AK. (2026) <https://biorender.com/cnobk3v>.

mitochondrial mobilization,¹⁶ whereas Cx43 deficiency recruits AMPK to mitochondria and suppresses transfer; pharmacologic AMPK inhibition restores mitochondrial exchange and accelerates stromal regeneration and hematopoietic recovery after injury.¹⁸ A schematic of Cx43 functions in HSPCs is presented in Figure 1.

However, major limitations remain. Efficient uptake and durable integration of donor mitochondria vary across cellular contexts and depend on metabolic compatibility between donor and recipient cells. Mitochondrial heteroplasmy and mitonuclear incompatibility also pose potential risks that may compromise long-term metabolic stability following transplantation. In addition, mitochondrial delivery can activate innate immune pathways, as cytosolic mitochondrial DNA engages cGAS STING signaling and induces type I interferons and inflammatory cytokines.^{84,85} Addressing these challenges will be essential for translating mitochondrial transfer strategies into safe and effective therapies for hematopoietic regeneration.

HSPC autonomous roles of Cx43

HSPC cytoplasmic and nuclear roles of connexins The C-terminal tail of Cx43 interacts with kinases, phosphatases, and transcriptional regulators, modulating their activity and downstream signaling pathways.⁸⁶ Acting as a multifunctional signaling scaffold, the C-terminus regulates pathways including MAPK,⁸⁶ PI3K/Akt,⁸⁶ and β -catenin.⁸⁷ Through these interactions, connexins interact with focal adhesion molecules,⁸⁸⁻⁹⁰ 14-3-3 molecular scaffolds,⁹¹ actin-dependent cytoskeletal dynamics,³⁵ and broader signal integration platforms⁸⁷ independently of GJ coupling. These nonjunctional roles expand the functional repertoire of connexins beyond intercellular continuity and contribute to the regulation of lineage commitment, stress responses, and proliferative capacity in HSPCs.⁹²⁻⁹⁴

Through its scaffolding functions, Cx43 influences local transcriptional programs by regulating the activity and subcellular localization of transcription factors⁸⁷ and chromatin-associated regulators.^{95,96} Of note, Cx43 C-terminal fragments can translocate to the nucleus, where they directly participate in transcriptional regulation of adhesion molecules by interacting with nuclear proteins and transcriptional complexes, shaping gene expression programs that govern cell fate.⁹⁵ In parallel, Cx43 regulates cytoskeletal organization and mechanotransduction through interactions with actin-binding proteins³⁵ and microtubule-associated complexes,^{97,98} coupling extracellular cues to intracellular structural and transcriptional responses. These mechanisms enable connexins to regulate not only cell-to-cell communication but also autonomous intracellular programs that control quiescence, activation, and differentiation trajectories. These redefine connexins as intracellular signaling organizers rather than solely intercellular conduits, broadening their role in hematopoietic regulation.

Regulation of HSC self-renewal, quiescence, and differentiation

Deletion of Cx43 in the hematopoietic system selectively impairs short-term repopulating activity under stress, such as following 5-fluorouracil-induced myeloablation, while sparing long-term competitive steady-state repopulating capacity.^{15,40} Hence, connexin-mediated communication is dispensable for baseline HSC maintenance but becomes critical during regenerative hematopoiesis, when metabolic and oxidative demands are heightened.

Cx43 is localized to both the inner and outer mitochondrial membranes in HSPCs, fundamentally reframing the role of connexins in hematopoiesis.¹⁸ Quiescent HSCs lacking Cx43 display increased mitochondrial AMPK activation and elevated ADP-to-ATP ratio, consistent with impaired bioenergetics.¹⁸ Reduced mitochondrial respiration and dysregulated expression of tricarboxylic acid cycle cycle genes (notably OGDH)

indicate Cx43 deficiency compromises oxidative metabolism and ability to meet energetic demands. Evidence for direct Cx43 AMPK interaction supports a model in which mitochondrial Cx43 helps set the threshold for AMPK activation, balancing energy supply and proliferative decisions. Cx43 acts as a rheostat that preserves regenerative potential by preventing chronic energy stress and untimely AMPK-driven proliferative suppression or catabolic program induction.

HSC polarity and spatial organization within the BM niche HSCs exhibit defined cell polarity that is regulated by upstream interactions with BM niche cells.^{99,100} Polarity enables HSCs to spatially organize adhesion receptors, signaling complexes, and organelles, facilitating directional communication from and to the BM niche. In HSCs, polarized distribution of adhesion receptors and intracellular structures regulates asymmetric division, migration, and lineage commitment.¹⁰¹⁻¹⁰³ HSC polarity is regulated in part by Cdc42-dependent signaling, which controls cytoskeletal organization and polarity establishment.¹⁰²⁻¹⁰⁴ Downstream of Cdc42, polarity pathways involving the Scribble complex contribute to stabilization of polarized membrane domains and adhesion signaling required for HSC-niche interactions.¹⁰⁴ Conversely, atypical protein kinase C polarity proteins do not significantly impair HSC maintenance,¹⁰⁵ indicating distinct polarity modules differentially regulating stem cell function. Indeed, BM endothelial Jagged2 regulates HSC polarity and fate through both cell-extrinsic and cell-intrinsic Notch signaling.¹⁰⁶ Cx43-mediated GJs support the establishment and maintenance of polarity by coordinating cell-to-cell communication, adhesion, and cytoskeletal organization.^{37,107} Recent studies indicate mitochondrial positioning and polarized mitochondrial inheritance influence HSC fate by regulating metabolic state, ROS levels, and asymmetric division outcomes.^{108,109} Within polarized membrane domains, Cx43 controls ROS transfer, mitochondrial dynamics, Ca^{2+} levels, and mitochondrial distribution amenable to transfer to the surrounding BM microenvironment.^{18,110}

Dysregulated mitochondrial dynamics and calcium handling During replicative stress, Cx43 deficiency disrupts mitochondrial dynamics in cycling HSCs, impairing Mfn2-dependent fusion¹¹¹ and increased Drp1-mediated fission,¹⁰⁹ leading to mitochondrial fragmentation, hypermotility, and enhanced mitophagy.¹⁸ These changes are accompanied by elevated mitochondrial ROS and reduced OXPHOS, impairing mitochondrial function and diminishing HSC regenerative capacity. Interventions blocking excessive fission, such as dominant-negative Drp1-K38A, or restoring mitochondrial complement via ex vivo functional mitochondrial supplementation rescue metabolic defects and in vivo HSC repopulation.¹⁸ In the highly enriched calcium environment of BM, calcium is a major regulator of HSC activity.¹¹² Mitochondrial calcium regulates mitochondrial motility and dynamics in HSC.¹¹³ Cx43-deficient HSCs exhibit elevated mitochondrial Ca^{2+} in both quiescent and cycling states despite unchanged cytosolic Ca^{2+} levels.¹⁸ Cx43 facilitates mitochondrial Ca^{2+} efflux,¹⁸ directly or in coordination with other mitochondrial calcium transporters, maintaining fusion-fission balance. Interestingly, minimal expression of canonical mitochondrial $\text{Ca}^{2+}/\text{Na}^{+}$ and $\text{Ca}^{2+}/\text{Li}^{+}$ exchanger¹¹⁴ in HSCs¹¹⁵ underscores Cx43 as the primary regulator of mitochondrial calcium in these

cells. Multicompartmental functions of Cx43 in HSPCs are presented in [Figure 2](#).

Mitochondrial Cx43 integrates bioenergetics, mitochondrial dynamics, and K^{+} , H^{+} , Ca^{2+} , and ATP homeostasis, preserving HSC regenerative capacity during activation and bone loss.^{18,116} By supporting OXPHOS, limiting basal AMPK activation, and maintaining fusion-fission balance during replication by preventing mitochondrial Ca^{2+} overload, Cx43 safeguards self-renewal. Its interaction with AMPK and localization to both mitochondrial membranes suggest multiple control points over enzymatic activity, ion flux, and membrane dynamics.¹⁸ Mitochondrial Cx43 supports HSPC bioenergetics, protecting cells from oxidative stress and apoptosis. Concurrently, HSPC Cx43 facilitates metabolic niche restoration through indirect mitochondrial transfer, linking progenitor adaptation to extrinsic niche repair. This integrated mechanism resolves longstanding ambiguities in earlier literature, where defects in stress hematopoiesis were attributed to impaired adhesion, soluble signaling, or GJ-mediated metabolite transfer. Therefore, Cx43 acts as a regulator of upstream mitochondrial remodeling and dynamics, providing a framework for understanding progenitor-niche interplay during hematopoietic regeneration.^{16,18}

BM HSC niche connexins in hematopoietic disease

BM failure, myelodysplastic syndromes, and myelofibrosis

BM failure syndromes, myelodysplastic syndromes, and myelofibrosis, characterized by chronic inflammation, stromal dysfunction, extracellular matrix remodeling, and ineffective hematopoiesis, reflect disruption of hematopoietic niche integrity.^{100,117} Proinflammatory cytokines, tumor necrosis factor α , interferon gamma, transforming growth factor β , and interleukin-1 β (IL-1 β) drive mesenchymal stromal activation, fibrotic remodeling, and loss of niche factors such as CXCL12 and Kit ligand, impairing HSPC maintenance, retention, and regeneration.^{118,119} In myelodysplastic syndromes, dysregulated HIF-1 α signaling, DRP1-dependent mitochondrial fission, and mitochondrial dysfunction drive ineffective hematopoiesis and clonal evolution through structural abnormalities, iron overload, and increased oxidative stress.^{120,121} These pathological changes converge on disrupted communication between HSPCs and the niche, implicating altered connexin-mediated GJ coupling as a driver of disease. Cx43-dependent intercellular communication coordinates metabolic cooperation, redox buffering, and stress adaptation between HSCs and stroma, whereas disruption impairs hematopoietic regeneration.^{15,40} In myelofibrosis, inflammatory signaling from malignant clones induces stromal activation, matrix stiffening, and fibrotic niche remodeling that physically and biochemically disrupts GJ formation and functional coupling, amplifying niche exclusion and ineffective hematopoiesis.¹¹⁹ These findings support a model in which connexin dysfunction is not merely secondary to inflammation or fibrosis but actively contributes to marrow failure pathogenesis by fragmenting niche communication networks, amplifying oxidative stress, promoting stem cell exhaustion, and enabling clonal dominance.¹⁷

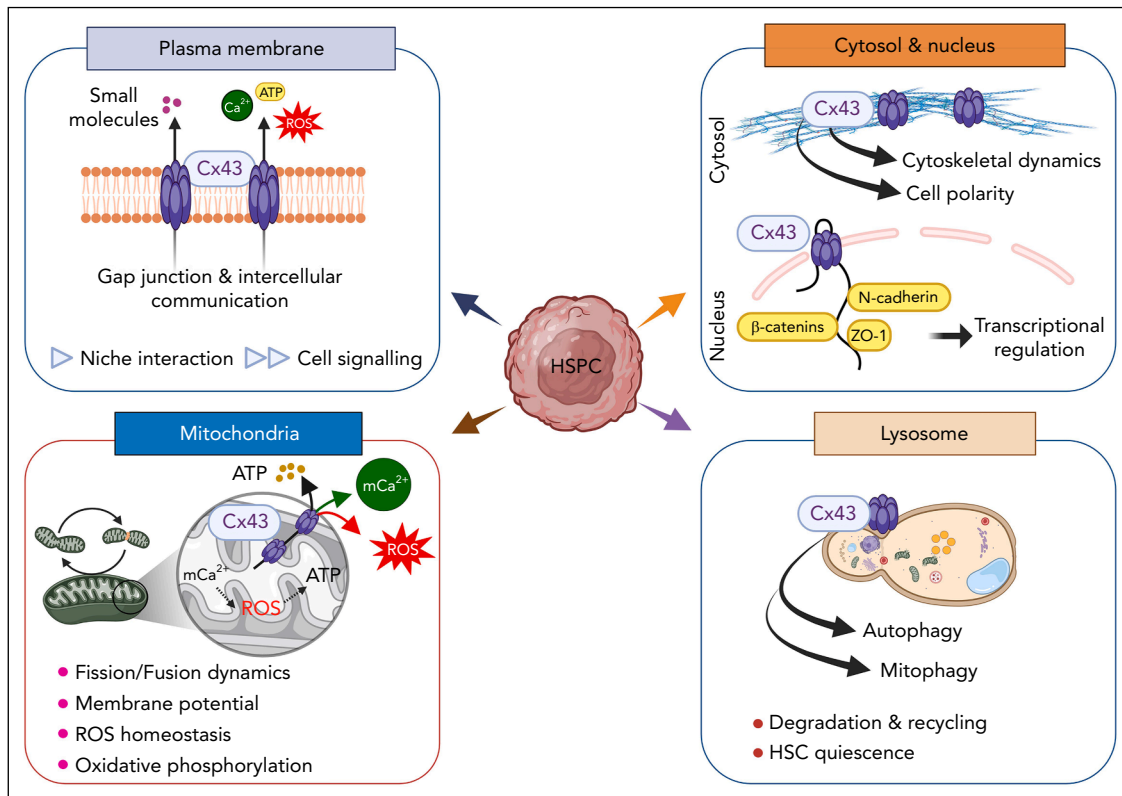


Figure 2. Multicompartmental functions of Cx43 in HSPCs. Cx43 operates across multiple cellular compartments to coordinate hematopoietic stress responses. At the plasma membrane, Cx43 GJs enable intercellular coupling and facilitate ROS buffering by transferring oxidative burden from HSPCs to stromal cells, whereas undocked hemichannels contribute to paracrine signaling. In the cytoplasm, Cx43 interacts with cytoskeletal and polarity complexes. Nuclear-associated pools of Cx43 interact with junctional and adhesion proteins, including β -catenin, N-cadherin, and ZO-1, linking Cx43 to transcriptional regulatory programs. At the mitochondrial level, discrete Cx43 pools localize to the inner and outer mitochondrial membranes, where they regulate mitochondrial Ca^{2+} handling, membrane potential, OXPHOS, ATP production, and ROS homeostasis. Lysosomal trafficking of Cx43 engages autophagy and mitophagy pathways, supporting organelle turnover and metabolic homeostasis. These compartment-specific functions position Cx43 as a central metabolic checkpoint during stress hematopoiesis. Figure created with BioRender.com. Singh AK. (2026) <https://biorender.com/dww7jtq>.

Leukemia and chemoresistance

Connexins coordinate bidirectional signaling between leukemic cells and the stroma, integrating inflammatory cues, redox buffering, metabolic exchange, and stress adaptation, influencing disease progression and therapeutic response.¹²²⁻¹²⁶

Leukemia exploits connexin-mediated GJs and hemichannels to support metabolic coupling, proliferation, survival, stemness, and chemoresistance while also selectively down-regulating connexins to evade prodeath signaling.^{124,125,127,128} Acute myeloid leukemia (AML) blasts show the highest expression of Cx43 and Cx45, activating MAPK signaling and IL-17, tumor necrosis factor α , and interferon gamma secretion, reinforcing protumorigenic microenvironments.¹²⁹ Additionally, connexins Cx25, Cx31.9, Cx40, and Cx59 are upregulated in AML, acute lymphoblastic leukemia, and CD34⁺ progenitors, and selective inhibition of Cx25 reduces proliferation, restoring chemosensitivity.^{124,129}

Leukemic stem and progenitors co-opt the BM niche to withstand cytotoxicity, with connexin-based intercellular communication critically supporting malignancy networks. AML BM-MSCs highly express Cx43 and Cx45 and other connexins, enabling GJ-mediated transfer of secondary messengers, metabolites, and redox equivalents, which buffer ROS, redistribute calcium, and sustain leukemic stemness under

cytotoxicity.¹²⁴ In myeloma, elevated tumor and BM Cx43 promotes adhesion, migration, and secretion of IL-6, SDF-1 α , and IL-10, reinforcing resistance.^{130,131}

Beyond small molecule exchange, leukemia exploits connexin-regulated TNTs and contact-dependent endocytosis to acquire intact, respiration-competent mitochondria from MSCs and other niche cells. This transfer supports prosurvival cytokine secretion, metabolic rewiring, increased ATP production, OXPHOS, and glutathione-dependent antioxidant defenses, reinforcing chemoresistance and minimal residual disease.^{123,132-134} Metabolic stress in leukemic progenitors, including glucose deprivation, NOX2-derived superoxide, and low ATP-to-AMP ratio, activates AMPK and FIS1 signaling and promotes mitochondrial fission, mitophagy, cytoskeletal remodeling, macropinocytosis, and TNT formation,¹³⁴⁻¹³⁶ whereas stromal PGC1 α -dependent mitochondrial biogenesis sustains transferable organelles.¹³⁷ Hypoxic BM conditions further enhance transfer through HIF-1 α dependent upregulation of Cx43, increasing GJ permeability and TNT docking, thereby enabling malignant cell adaptation to nutrient and oxygen limitation. Elevated CD38 expression in hematologic malignancies also regulates mitochondrial trafficking,¹³³ although its role in extracellular vesicle-mediated transport remains unclear. Malignant cells can additionally acquire

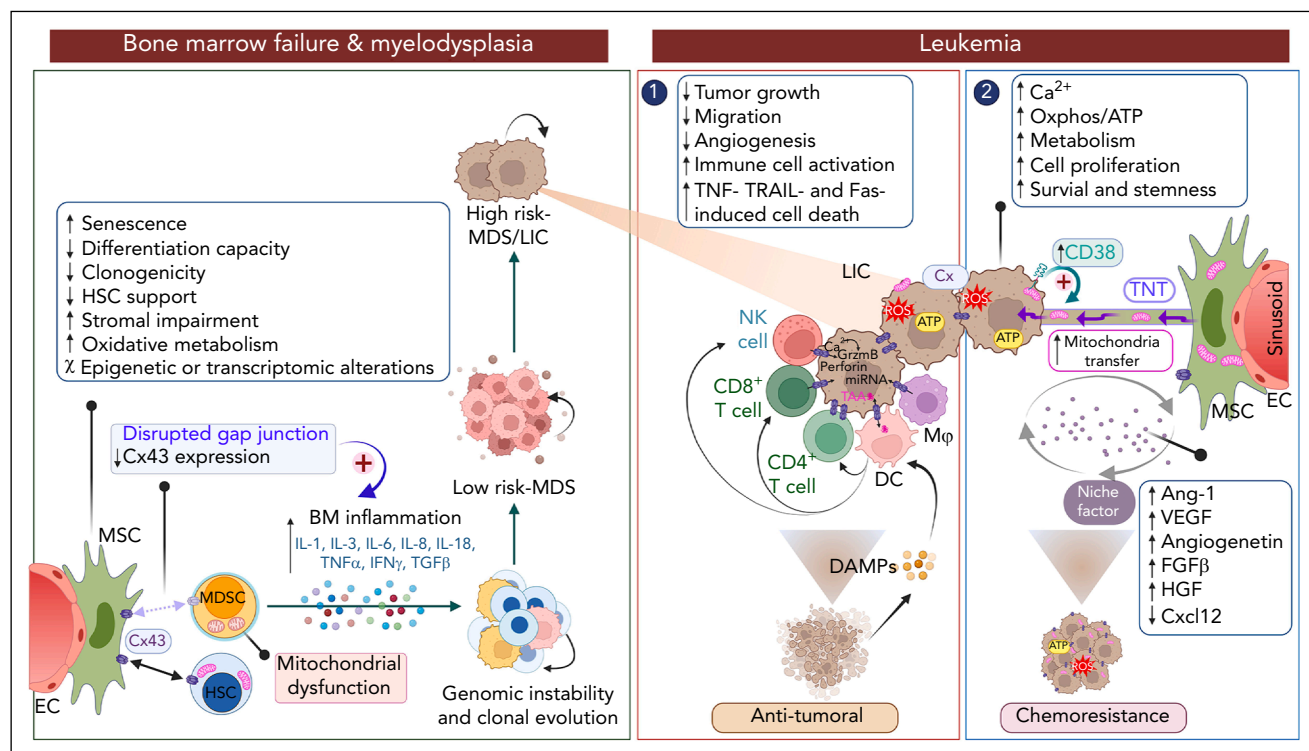


Figure 3. Pathologic rewiring of connexin networks in hematologic disease. Dysregulated connexin signaling contributes to malignant transformation, immune modulation, and impaired regeneration across hematologic disorders. In leukemia, malignant cells exploit connexin-mediated GJs and hemichannel activity to enhance metabolic coupling, mitochondrial transfer, survival, stemness, and chemoresistance while selectively downregulating connexins in specific contexts, aiming to evade prodeath signaling. In BM failure syndromes and MDS, disrupted GJ coupling, chronic inflammation, and fibrosis impair stress sensing and regenerative capacity. Mitochondrial dysfunction in MDS drives ineffective hematopoiesis and clonal evolution through structural abnormalities, iron overload, and increased oxidative stress in stem cells. Within the tumor microenvironment, connexins, particularly Cx43, exert complex, context-dependent effects. (1) Cx43 GJs and hemichannels facilitate transfer of tumor-associated antigens and DAMPs from dying tumor cells to DCs, promoting DC activation, CD8⁺ and CD4⁺ T cell priming, NK cell activation, and macrophage-mediated transfer of tumor-suppressive microRNAs. (2) Conversely, GJs between tumor cells and stromal cells promote tumor progression. In the leukemic BM niche, elevated Cx43 expression in LICs, blasts, and stromal cells facilitates the exchange of survival signals. Oxidative stress induces mitochondrial transfer from stromal cells via TNTs, a process enhanced by tumor cell CD38. Transferred mitochondria reprogram leukemic bioenergetics and metabolism, promoting leukemogenesis and chemoresistance. Stromal signaling through Ang-1/Tie2, VEGF, angiogenin, FGFβ, and CXCL12 further protects leukemic cells from chemotherapy. DAMP, danger-associated molecular patterns; DC, dendritic cell; EC, endothelial cell; FGFβ, basic fibroblast growth factor; IFN-γ, interferon gamma; LIC, leukemia-initiating cell; MDS, myelodysplastic syndromes; MDSC, myeloid-derived suppressor cell; NK, natural killer; TGF-β, transforming growth factor β; TNF-α, tumor necrosis factor α; TRAIL, tumor necrosis factor-related apoptosis-inducing ligand; VEGF, vascular endothelial growth factor. Figure created with [BioRender.com](https://biorender.com). Singh AK. (2026) <https://biorender.com/x0ailov>.

mitochondria from T cells, macrophages, and other immune cells, enhancing tumor bioenergetics while returning dysfunctional mitochondria containing mutated mitochondrial DNA, promoting immune metabolic impairment, senescence, and defective antigen presentation.^{84,138}

Targeting Cx43 offers a nuanced therapeutic opportunity, as its functional impact is context dependent. Although it coordinates metabolic coupling that supports niche repair, bidirectional mitochondrial exchange between stromal cells, leukemic blasts, and immune cells may promote tumor survival, immune evasion, and therapy resistance. Selective disruption of pathological leukemic-stromal connexin interactions, preserving normal hematopoietic support, could weaken niche-mediated chemoresistance, sensitizing leukemia to treatment.¹²⁵ Conversely, selective uncoupling may shield malignant clones from niche-derived growth restraints or differentiation cues,¹²⁴ implying that modulating Cx43 activity requires precise, context-specific strategies. Additionally, connexin-dependent communication can redistribute mitochondrial ROS, calcium, and other metabolites away from cancer cells, diluting chemotherapy or targeted therapy damage. A schematic

illustrating how connexin networks contribute to pathological rewiring of the BM niche is presented in [Figure 3](#).

The context-dependent role of connexin networks and their dysregulation in health and disease is an exciting field. Currently, one of the main challenges is that precise mechanisms of the adaptation of connexins in disease have not been characterized. Definitive mechanistic demonstration on how connexins drive marrow failure, clonal evolution, leukemic niche remodeling, or chemoresistance will require cell-type-specific genetic models and in vivo functional assays to establish causality.

Translational potential, clinical perspective, and therapeutic opportunities

Enhancing endogenous repair and regeneration after cytotoxic injury

Given that Cx43 expression in marrow is essential for hematopoietic recovery after chemotherapy, therapeutically boosting connexins may accelerate hematologic recovery.

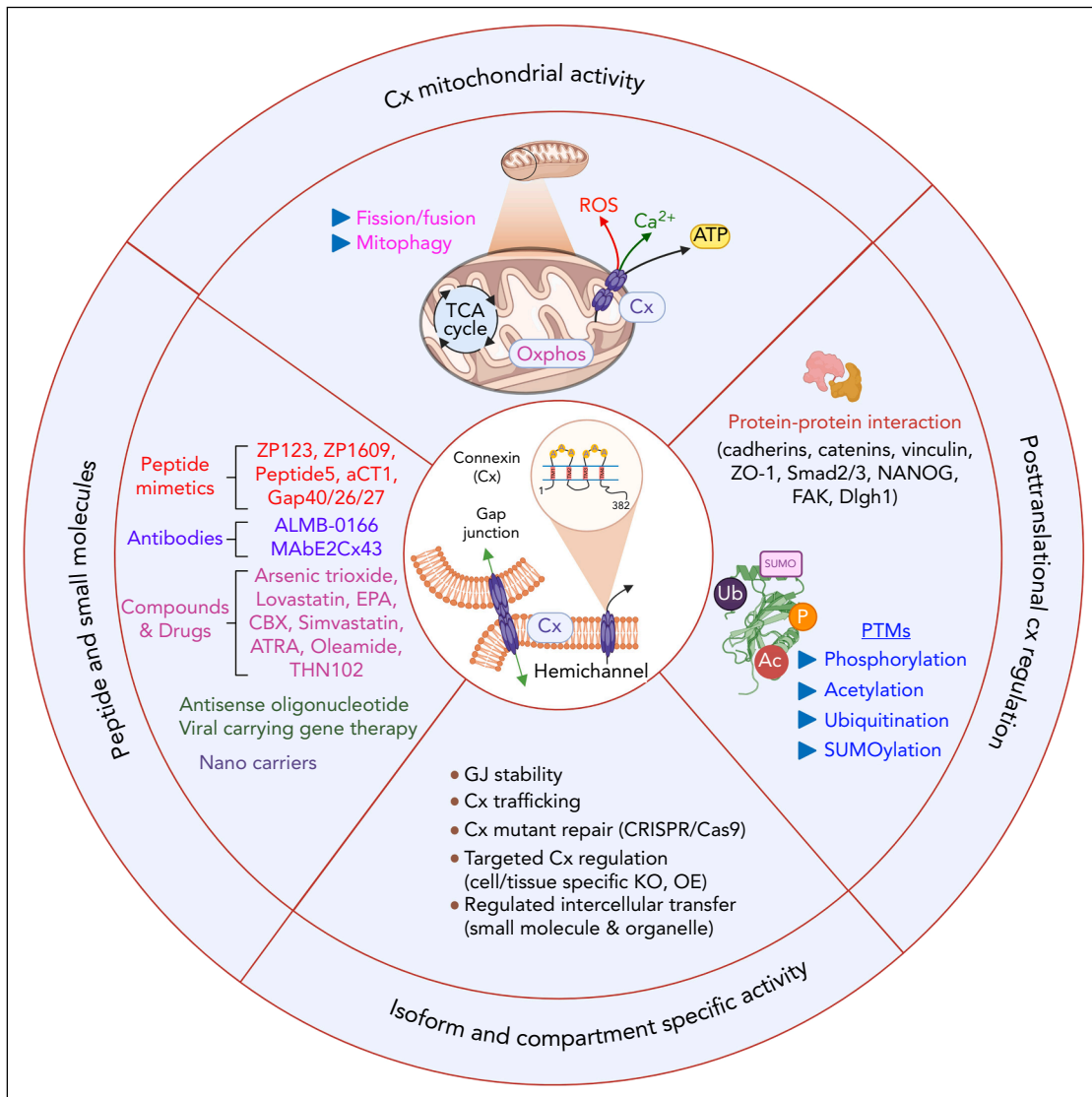


Figure 4. Translational opportunities targeting connexins in hematopoiesis and regenerative medicine. Connexins, particularly Cx43, function across plasma membrane GJs, hemichannels, cytoplasm, nucleus, and mitochondria to regulate HSC homeostasis, stress responses, redox balance, and bioenergetics. Four major translational domains are highlighted (clockwise from top). Mitochondrial and metabolic targeting: modulation of mitochondrial Cx43 influences fusion-fission dynamics, mitophagy, calcium handling, OXPHOS, ATP production, and ROS homeostasis, preserving HSPC metabolic fitness under replicative and inflammatory stress. Posttranslational regulation: phosphorylation, acetylation, ubiquitination, and SUMOylation control connexin trafficking, stability, and protein interactions, including interactions with cytoskeletal proteins, FAK, and scaffold complexes that regulate tumor growth and migration. Isoform- and compartment-specific modulation: selective targeting of connexins (Cx43, Cx37, Cx45, and Cx32) using knockdown, overexpression, or gene-editing strategies (eg, CRISPR-Cas9) enables correction of disease-associated mutations, regulation of GJ stability, and fine-tuning of intercellular and intracellular communication. Pharmacologic and biologic interventions: small molecules, antibodies, mimetic peptides, antisense oligonucleotides, and nanocarrier-based delivery systems can modulate GJs, hemichannels, or mitochondrial Cx43 to enhance protective signaling during stress or inhibit pathological communication in disease. These strategies position connexins as actionable therapeutic targets for modulating hematopoietic function, regeneration, and disease. FAK, focal adhesion kinase; KO, knockout; OE, overexpression; PTMs, posttranslational modifications. Figure created with BioRender.com. Singh AK. (2026) <https://biorender.com/1y1342g>.

In mice conditionally deficient for Cx43, administration of 5-fluorouracil impairs blood count recovery (neutrophils, erythrocytes, and platelets), progenitor frequency (approximately fivefold to eightfold reduction), and marrow cellularity, demonstrating Cx43 requirement for regeneration.^{15,40} These deficiencies, not present at baseline, emphasize the critical role of connexin coupling in regenerative, but not in steady-state hematopoiesis. Furthermore, BM regeneration after transplantation requires mitochondrial transfer from donor Cx43-expressing hematopoietic cells to stromal cells.¹⁶ This mitochondrial transfer reciprocally supports niche energy

metabolism, sustaining engraftment efficiency, supporting that Cx43-dependent mitochondrial handoff is foundational for marrow recovery.

This points to a regenerative paradigm. HSCs not only receive signals from the niche but may dispatch metabolic cargo back to stromal cells, thus co-optimizing niche function sustaining hematopoiesis. Therapeutically, supporting or enhancing mitochondrial transfer and connexin coupling may accelerate recovery after transplantation, chemotherapy, or irradiation.

Selectively disrupting pathological niche support

Conversely, selectively inhibiting connexin coupling between malignant and niche cells could strip leukemic clones of metabolic support, enhancing chemosensitivity. Disrupting connexin/GJ interactions with agents like carbenoxolone reduces niche-induced chemoresistance, increases apoptosis in AML, and synergizes with cytotoxic drugs without adversely affecting normal progenitors.¹²⁴ Possible tools include connexin mimetic peptides (eg, Gap26, Gap27, and L2/Gap19)¹³⁹⁻¹⁴² that block GJ and hemichannel communication,¹⁴³ isoform-specific antibodies targeting connexin hemichannels,¹⁴⁴ or small-molecule inhibitors targeting specific connexin isoforms or their docking interfaces to alter tumor-stroma interactions.¹⁴⁵ Precision targeting is key in this strategy, and because connexins are ubiquitous, therapies must avoid excessive collateral damage to healthy BM cells.

Bioengineering and ex vivo HSC expansion

Because mitochondrial transfer regulated via Cx43 is crucial to niche repair and HSPC-stromal metabolic exchange, modulating this process offers a therapeutic axis. Enhancing mitochondrial donation or forcing mitochondrial transfer into ex vivo manipulated HSC may accelerate regeneration after myeloablative injury by boosting stromal recovery and hematopoietic reconstitution,¹⁸ whereas inhibiting mitochondrial transfer into malignant pathologic stromal compartments may impair pathologic plasticity, leukemia support, and metabolic reprogramming.¹²³

Ex vivo systems that mimic physiologic connexin networks, such as MSC cocultures that recapitulate stromal support and 3-dimensional biomimetic scaffolds engineered to present niche-like physical and biochemical cues, may better preserve HSC function and support expansion compared with conventional cultures.¹⁴⁶ For example, MSC-HSPC coculture platforms enhance progenitor proliferation and maintenance ex vivo, alginate hydrogel scaffolds with niche signals sustain HSC quiescence, and biomimetic micro-niche structures significantly expand primitive HSC subsets, illustrating the promise of bioengineered approaches for improving HSC expansion and transplantation outcomes.¹⁴⁷⁻¹⁵⁰ Scaffold-based cocultures or microfluidic devices engineered to allow controlled GJ coupling or cocultures with stromal cells expressing defined connexin complements may improve HSC expansion and transplantation outcomes.^{151,152}

Biomarkers

Connexins may serve as biomarkers of marrow health and regenerative potential. Differential expression of Cx32, Cx43, and Cx45 has been observed across AML subsets, with profiles associated with cell maturation and signaling phenotypes that could inform disease classification.¹²⁹ Moreover, clinical studies show Cx32 and Cx43 mRNA levels correlate with hematopoietic parameters such as CD34 expression in AML samples, suggesting that connexin expression profiles might carry prognostic or biomarker value.¹⁵³ Additionally, preclinical models demonstrate that Cx43 expression within the marrow niche is required for efficient hematopoietic recovery after cytotoxic injury and regulates HSC homing and engraftment, implying that connexin expression or coupling function in patient marrow samples (before/after transplantation) might predict engraftment success or graft failure risk.^{14,40}

In vivo therapeutic targeting

Connexin-based therapies could serve as adjuncts to transplantation, chemotherapy, and immunotherapy by accelerating hematopoietic recovery, protecting normal HSCs, or disrupting malignant niche support. Combinatorial strategies, such as connexin activation with cytokines or inhibition with targeted therapies, offer promise, linking connexin activation with cytokine support to increase chemosensitivity,¹²⁵ or connexin inhibition with targeted leukemia therapies.¹²⁴

Translational challenges include tissue specificity, temporal control, and safety. Potential approaches include small molecules, biologics, antibodies, and gene-based strategies. GJ openers (eg, ALMB-168¹⁵⁴ and antiarrhythmic peptide 10¹⁵⁵) or not-so-specific Cx43 inhibitors (eg, rotigaptide,¹⁵⁶ carbenoxolone, and related glycyrrhizin derivatives⁴²) may be repurposed to enhance coupling during regeneration or block coupling in malignancy.¹⁵⁷⁻¹⁵⁹ However, achieving marrow-specific activity and avoiding cardiac or vascular off-target effects remain challenges. Antibodies or engineered proteins targeting extracellular loops of connexins could selectively block or stabilize GJ or hemichannels in cell subsets (eg, leukemic clones), potentially reducing toxicity.¹⁶⁰ CRISPR/Cas9, RNAi, or viral gene delivery strategies offer precise connexin function restoration in regenerative contexts or selectively silence connexins in malignant clones, providing a promising but largely preclinical therapeutic avenue. A depiction of translational opportunities targeting connexins in hematopoiesis and regenerative medicine is presented in Figure 4.

Conclusions

Connexins are central regulators in the BM microenvironment, integrating intercellular communication with intrinsic metabolic control. Beyond conduits, connexins, particularly Cx43, coordinate redox buffering, mitochondrial function, and organelle trafficking, processes essential for HSC survival and regenerative hematopoiesis. Pathway disruption impairs recovery, accelerates stem cell aging, and promotes malignant niche remodeling and chemoresistance. Recognition of mitochondrial Cx43 as a metabolic checkpoint reframes GJ biology in hematopoiesis, reconciling the paradox that connexins are dispensable at steady state yet indispensable during regeneration. Discovery of Cx43-dependent mitochondrial transfer from HSPCs to stroma further establishes a bidirectional niche-repair paradigm. These insights position connexins as powerful but complex therapeutic targets; enhancing physiological connexin function may accelerate recovery after injury or transplantation, whereas disrupting pathological coupling may sensitize leukemia to therapy. Progress will require isoform- and compartment-specific dissection, structural definition of mitochondrial connexin function, and development of temporal- and tissue-specific modulators to translate connexin biology into hematologic therapeutics.

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Footnotes

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